



OTC Derivatives Reporting and the Classification Frontier

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Modules 1-2 turn noisy raw trades into structured inputs.

Parsing, taxonomy routing, ANNA-DSB lookup, and end-to-end integration

Capability	What I implemented	Why it matters
Fault-tolerant parsing	Missing / null / malformed values still return a ParsedTrade result	Bad rows stay visible instead of disappearing
Taxonomy routing	Conventional derivatives vs. novel event contracts	T026-T028 and T033 avoid false normalisation
UPI template matching	ANNA-DSB product definitions + filename alias mapping	Trades reach the correct DSB template family
Codeset validation	Currency, rate, enum, and delivery-type checks	Hard failures are separated from soft warnings
Pipeline integration	28 provided trades + 5 additional trades in one run	One complete report for M3 and dashboard

Representative outcomes

Evidence from the final report

T005

LIBOR rate

FOUND + warning

T007

XAU currency

FOUND

T009

Invalid currency

INVALID_ATTRIBUTES

T026

Event contract

NO_PRODUCT_DEFINITION

Hard error != warning != taxonomy frontier

28

provided trades

05

additional trades

33

report rows

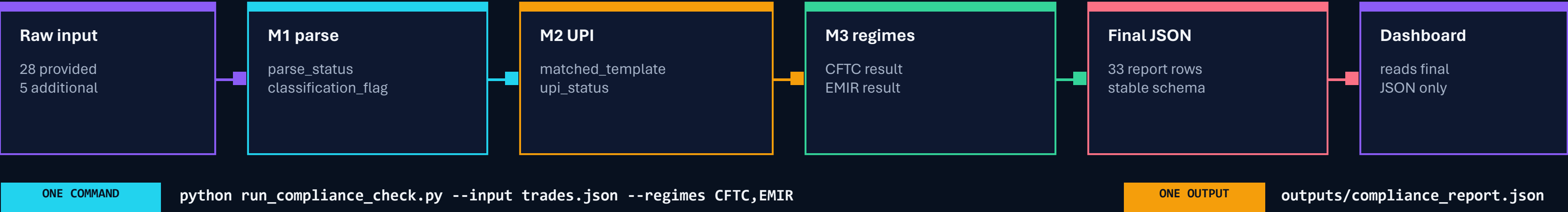
04

novel instruments

Takeaway: Modules 1 and 2 do not just clean data; they preserve edge cases and make later compliance logic interpretable.

One pipeline carries the same trade contract from raw JSON to the dashboard.

Final integration across parsing, UPI lookup, compliance checks, and visualisation



Shared interface contract

Field	Produced by	Consumed by	Why it matters
parse_status	Module 1	M3 + report writer	Keeps malformed rows visible
classification_flag	Module 1	M2 + M3	Routes event contracts away from false template matching
matched_template / upi_status	Module 2	M3 + report writer	Separates found, invalid, and no-taxonomy cases
regime_results	Module 3	Dashboard	One JSON structure drives the final visualisation

Takeaway: my role was not only to build two modules, but to make the whole project executable as one consistent reporting workflow.

Module 3: Validation + Event Contract Logic

What I implemented in Module 3

LEI	Format + ISO 7064 MOD 97–10 checksum
UTI	Length, namespace LEI, reporting LEI match, suffix format
EMIR extra fields	Collateral portfolio code, initial margin, variation margin
UPI handling	FOUND with null live UPI code is a note, not a hard error

Event contracts

ID	CFTC	EMIR
T026	CONDITIONAL	N/A
T027	N/A	N/A
T028	CONDITIONAL	N/A
T033	CONDITIONAL	N/A

Takeaway: Module 3 separates hard errors from warnings and keeps event contracts at the taxonomy frontier.

Five synthetic trades added to test clean controls, targeted failures and the event-contract frontier.

PYTHON OUTPUT SNAPSHOT

33

TOTAL TRADES

5

ADDITIONAL

2

CLEAN CONTROLS

2

INTENTIONAL ERRORS

1

EVENT VARIANT

Output-derived checks

- T029/T032 pass both regimes
- T030 fails UTI namespace
- T031 fails EMIR margin fields
- T033 creates a UPI taxonomy gap

Additional trades in compliance_report.json

Designed to separate clean controls from intentional failures

ID	Role	Intent	CFTC	EMIR	Output signal
T029	Clean	Rates control	COMPLIANT	COMPLIANT	Matched UPI + compliant
T030	Error	UTI namespace mismatch	NONCOMPLIANT	NONCOMPLIANT	Identifier validation triggered
T031	Error	Missing EMIR margin fields	COMPLIANT	NONCOMPLIANT	Jurisdiction-specific failure
T032	Clean	FX swap control	COMPLIANT	COMPLIANT	Matched UPI + compliant
T033	Frontier	FOMC event contract	CONDITIONAL	NOT_APPLICABLE	NO_PRODUCT_DEFINITION

Design principle

Targeted test cases expose validation issues without accidental LEI / UTI noise.

Module 4: Event Contracts & Engine Limits

Event contracts create a taxonomy gap: classification is necessary, but not enough for surveillance.

4A–4B Economic function + UPI gap

T026 Election

Mostly price discovery

T027 Offshore

Access + venue risk

T028 Regulatory

Clearer economic exposure

T033 FOMC

Monetary-policy event variant

Proposed UPI family

EventContract.Binary.FixedPayout

Key fields: event category, resolution source, payout rule, platform type, access model.

4C–4D Arbitrage + engine limits

Regime status counts for additional trades

CFTC

3

1

1

EMIR

2

2

1

COMPLIANT

NONCOMPLIANT

CONDITIONAL

NOT APPLICABLE

What the engine cannot see

VPN / offshore access • user geography • true economic exposure
market manipulation • insider risk • cross-platform positions

Takeaway

Current engine = first-stage classification tool, not a full event-contract surveillance system.

Module 3: Compliance Checker

Zhang
Yihan

CFTC + EMIR decision layer

Input

- MonetParsed trade from Module 1
- UPI lookup result from Module 2
- Raw trade fields for regime checks

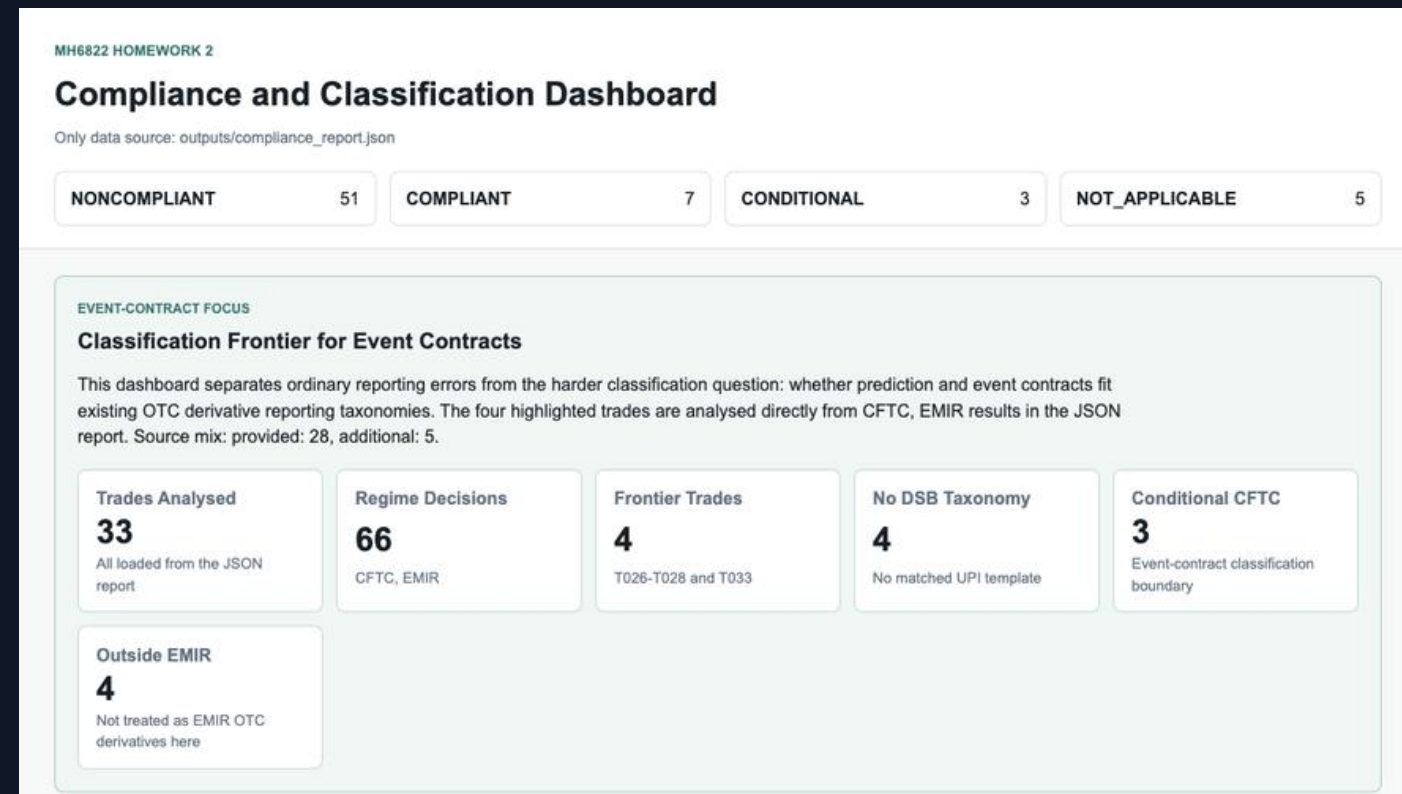
Output

- CFTC result
- EMIR result
- status + errors + warnings + notes

Status values: COMPLIANT / NONCOMPLIANT / CONDITIONAL / NOT_APPLICABLE

Module 5: Bonus Dashboard

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TOTAL TRADES

33

REGIME DECISIONS

66

FRONTIER TRADES

4

WHAT I BUILT

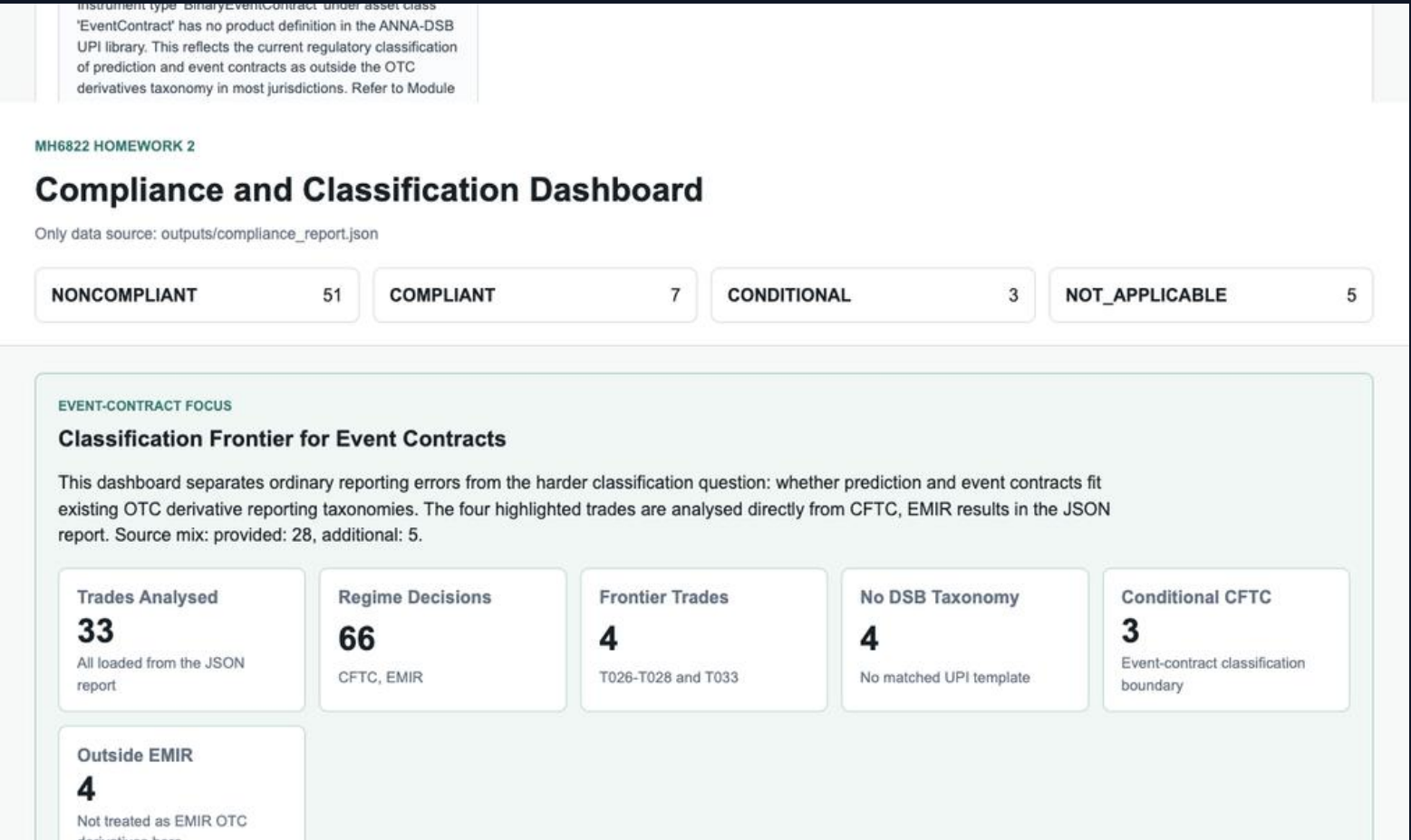
A dashboard generated only from the final compliance report

- Reads outputs/compliance_report.json only
- No manual hard-coding of trade outcomes
- Converts engine output into visual evidence
- Highlights event contracts separately from data errors

Takeaway: the dashboard is a presentation layer over the engine, not a separate source of truth.

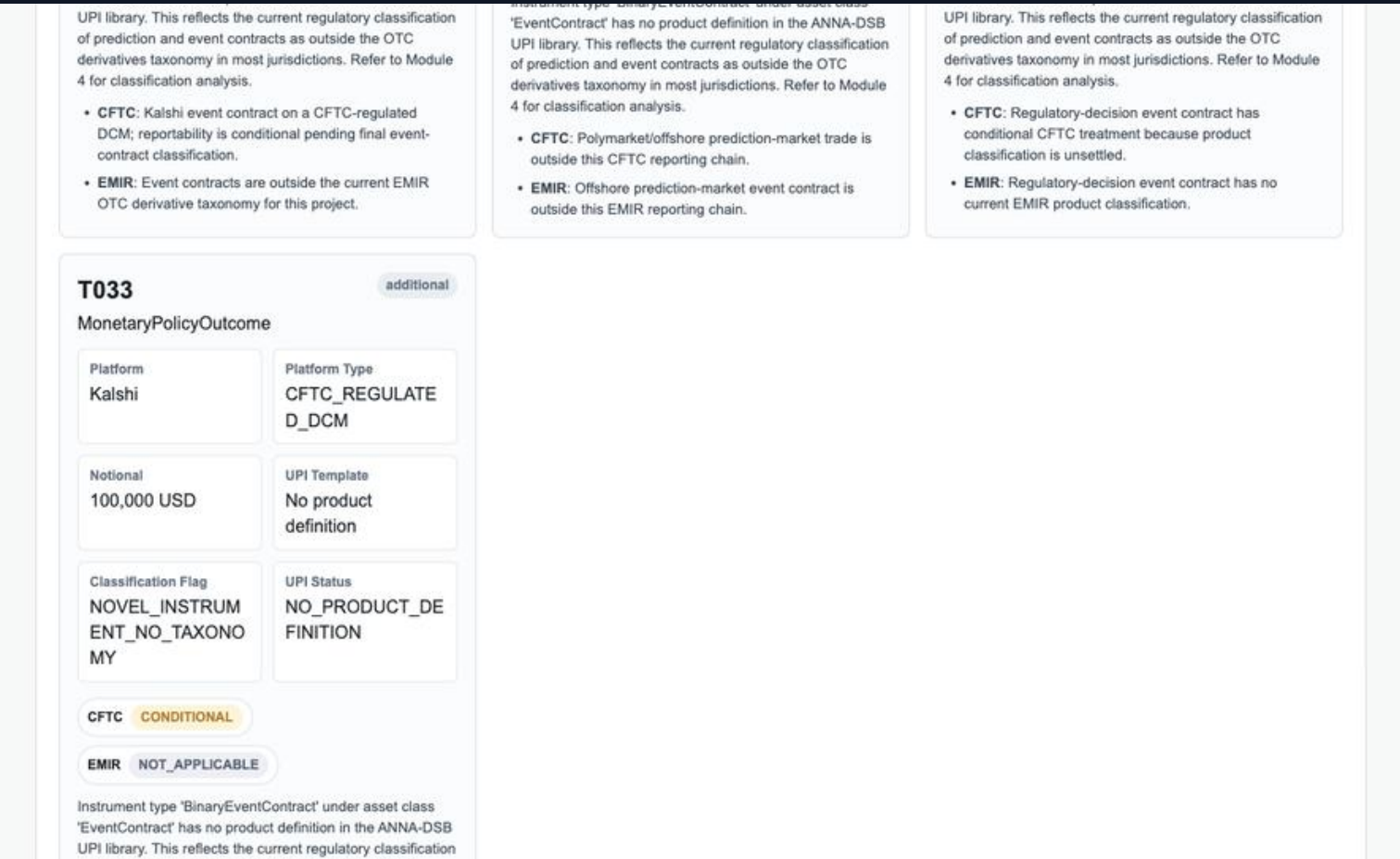
Dashboard Views: From Portfolio to Error Controls

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Portfolio compliance heatmap

Trade-by-trade status under CFTC and EMIR, with event-contract rows visually separated.



Error frequency + asset class breakdown

Repeated failures point to LEI, UTI and EMIR collateral / margin controls, not isolated one-off mistakes.

FOUR REQUIRED PANELS

Heatmap • Error frequency • Asset-class breakdown • Classification frontier

Classification Frontier: Event Contracts

PoliticalOutcome Platform Kalshi Platform Type CFTC_REGULATE D_DCM Notional 50,000 USD UPI Template No product definition Classification Flag NOVEL_INSTRUM ENT_NO_TAXONO MY UPI Status NO_PRODUCT_DE FINITION CFTC CONDITIONAL EMIR NOT_APPLICABLE Instrument type 'BinaryEventContract' under asset class 'EventContract' has no product definition in the ANNA-DSB UPI library. This reflects the current regulatory classification of prediction and event contracts as outside the OTC derivatives taxonomy in most jurisdictions. Refer to Module 4 for classification analysis. • CFTC: Kalshi event contract on a CFTC-regulated DCM; reportability is conditional pending final event-contract classification. • EMIR: Event contracts are outside the current EMIR OTC derivative taxonomy for this project.	MacroeconomicOutcome Platform Polymarket Platform Type DECENTRALISED_ BLOCKCHAIN_PLA TFORM Notional 200,000 USDC UPI Template No product definition Classification Flag NOVEL_INSTRUM ENT_NO_TAXONO MY UPI Status NO_PRODUCT_DE FINITION CFTC NOT_APPLICABLE EMIR NOT_APPLICABLE Instrument type 'BinaryEventContract' under asset class 'EventContract' has no product definition in the ANNA-DSB UPI library. This reflects the current regulatory classification of prediction and event contracts as outside the OTC derivatives taxonomy in most jurisdictions. Refer to Module 4 for classification analysis. • CFTC: Polymarket/offshore prediction-market trade is outside this CFTC reporting chain. • EMIR: Offshore prediction-market event contract is outside this EMIR reporting chain.	RegulatoryDecisionOutcome Platform Kalshi Platform Type CFTC_REGULATE D_DCM Notional 100,000 USD UPI Template No product definition Classification Flag NOVEL_INSTRUM ENT_NO_TAXONO MY UPI Status NO_PRODUCT_DE FINITION CFTC CONDITIONAL EMIR NOT_APPLICABLE Instrument type 'BinaryEventContract' under asset class 'EventContract' has no product definition in the ANNA-DSB UPI library. This reflects the current regulatory classification of prediction and event contracts as outside the OTC derivatives taxonomy in most jurisdictions. Refer to Module 4 for classification analysis. • CFTC: Regulatory-decision event contract has conditional CFTC treatment because product classification is unsettled. • EMIR: Regulatory-decision event contract has no current EMIR product classification.
T033 MonetaryPolicyOutcome Platform Platform Type		

T026, T027, T028 and T033 do not map to an ANNA-DSB UPI product definition.

KEY FINDING

Event contracts are a taxonomy boundary, not just bad data.

- Conventional failures: LEI, UTI, required fields
- Event contracts: no current DSB product template
- CFTC: conditional for regulated event platforms
- EMIR: outside the project taxonomy

Final message: the engine finds data failures and also exposes where product taxonomy stops.



THANK YOU !

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